

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MS 201 — Past Questions & Materials
by Renance

Course: MS 201
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

This study pack was compiled and restructured by Renance for MS 201 students at Federal University of Petroleum Resources, Effurun. It brings together 1 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app
Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 1 — MS 201 2024|2025

Level 200 • First Semester • PastQuestion

This source paper is a scanned document; its pages are reproduced below.

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES,
EFFURUN.
COLLEGE OF MARITIME AND OFFSHORE STUDIES
DEPARTMENT OF MARINE TRANSPORT AND LOGISTICS
MARITIME TECHNOLOGY (MS 201) -1st SEMESTER
EXAMINATION: 2024/2025 ACADEMIC YEAR

ANSWER QUESTION 1 PLUS ANY OTHER 4: TIME: 2 HRS 30 MINS.

Question 1

- a. You are tasked with designing a cathodic protection system for an 8000m^2 ship hull with a -0.7V metal potential, 0.5 coating breakdown factor, and $0.1\text{mA}/\text{m}^2$ steel current density. The ship owner prefers long flush-mounted anodes with -1.7V potential and 0.5m by 0.12m dimensions, operating in a $2.5\Omega\text{m}$ environment. Estimate the number of anodes required for the ship's protection (6.5 mrks)
- b. What causes torsion in a ship, and what orientation of the ship to the wave leads to this condition? (3 mrks)
- c. How does slamming contribute to the structural movement known as panting, and what causes its fluctuation? (3 mrks)
- d. What physical component of dry-docking directly causes an upward stress on the keel, and what are the resulting deformations? (3 mrks)
- e. Under what condition is racking most pronounced, and what dynamic force initiates it. (3 mrks)

Question 2

- a. Why does a ship experience increased bending moment during sagging? (3.5mrks)
- a. What causes the phenomenon known as slamming or pounding, and which part of the ship is primarily involved? (3 mrks)
- b. State two (2) differences between corrugated bulkhead and plane bulkheads. (3 mrks)
- c. What are the phases of ship design process? (3 mrks)

Question 3

- a. What structural effects occurs when a wave trough is at the ends of the ship? (3.5 mrks)
- b. In your understanding, how does the distinction between internal and external static forces influence structural decision-making in ship design? (3 mrks)
- c. What are intumescent coatings, and how do they provide fire resistance on board ships? (3 mrks)
- d. State 4 primary causes of corrosion on ships (3 mrks)

Question 4

- a. Describe the type of structural deformation that occurs during racking. (3.5 mrks)
- b. What motion of the ship enhances panting effect? (3 mrks)
- c. Why are transverse and longitudinal bulkheads considered critical structural elements in ship design, while minor bulkheads are not? (3 mrks)
- d. List 3 types of corrosion that are commonly encountered in marine environment. (3 mrks)

Question 5

- a. What happens to buoyancy when a wave crest is at mid-ships? (3.5 mrks)
- b. In what way do dynamic forces introduce complexities to ship structural design beyond those posed by static forces? (3 mrks)
- c. Transverse section of a ship is vulnerable to distortion due to water pressure, what structural feature prevents it? (3 mrks)
- d. Mention 3 types of materials commonly used as sacrificial anodes for cathodic protection in marine environment. (3 mrks)

Question 6

- a. What specific structural behaviors are observed in a ship during dry-docking due to static force? (3.5 mrks)
- b. Why is it essential to consider both static and dynamic forces together when evaluating a ship's long-term structural performance? (3 mrks)

- c. In Ship design, why is it important that a design philosophy accommodates a feedback mechanism, particularly when developing an innovative vessel type such as autonomous ships? (3 mrks)
- d. What are thermal loads in the context of ships, and how do they impact ship operations? (3 mrks)

Question 7

- a. What is the definition of floodable length in ship design, and how can it impact ship safety and operations? (3.5 mrks)
- b. What does the Design Spiral analogy demonstrate? (3 mrks)
- c. What are the six degrees of freedom referenced in ship movement? (3 mrks)
- d. What is a hatchway, and why are thicker insert plates fitted around hatch corners? (3 mrks)

Goodluck! Engr. N. E. Udosoh